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Properties of Materials
Lecture 5

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Lecture 3: Structure-Property-Processing Relationship, Structure Sensitive Properties, Structure Insensitive Properties, The Three-Part Relationship, Materials Property Web Sites, Materials Selection, Material Property - Charts: Modulus - Density, Mechanical Properties (2), Thermal Properties (2), A TOOL for Materials Selection, Structure Sensitivity, Structure Sensitivity in Metals, Controlling Structure, Case Study
Lecture 4: Availability of Materials, Case Study, Exam Question 2013, Example Questions, The World of Materials (2), Materials Selection in Design (2), Properties of Engineering Materials (3), Three-Way Relationship (2), Learning Objectives, Why Do We Need a Selection Methodology, Mechanical Properties (3), Thermal Properties (3), Trade-Offs Must Be Made, Materials Selection in Design (3), Materials Selection Process (1)
Lecture 5: Step 1 - Translation, Step 2 - Screening, Materials Selection Process (2), Step 3 - Ranking Materials, Example - Hydraulic Cylinder Tie Rods, Step 4 - Refining Selection, Concepts and Terminology (2)

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Availability of materials

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Step 1 – Translation

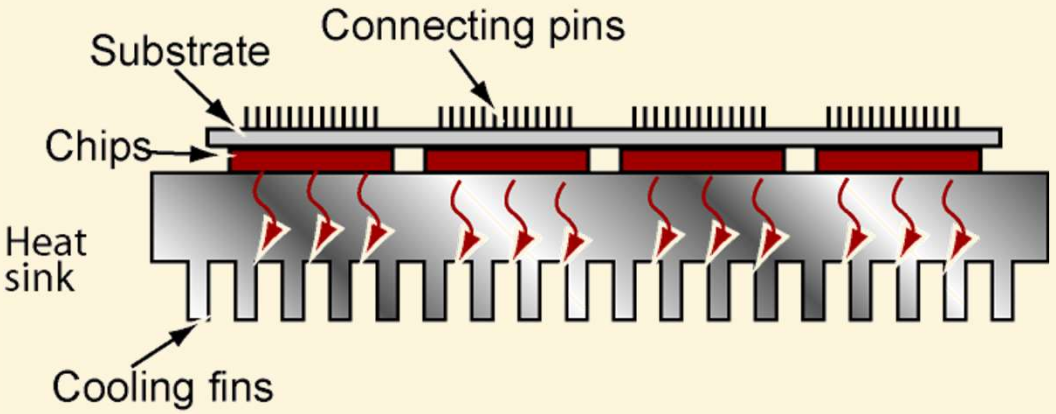
Design requirements

Function	• What does the component do?
Constraints	• What essential conditions must be met?
Objectives	• What needs to be maximised or minimised?
Free variables	• Which design variables are free?

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Translation example

- Heat sink for power electronics:
  - Describe function
  - Identify constraints
  - **Translate** into criteria



Function

- Absorb heat and convey away from chips

Constraints

- Operate at 200°C
- Be a good electrical insulator
- Be a good thermal conductor
- Lightweight

Image from: Ashby, M. F., (2011), *Materials Selection in Mechanical Design*

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Translation example (continued)

Constraints

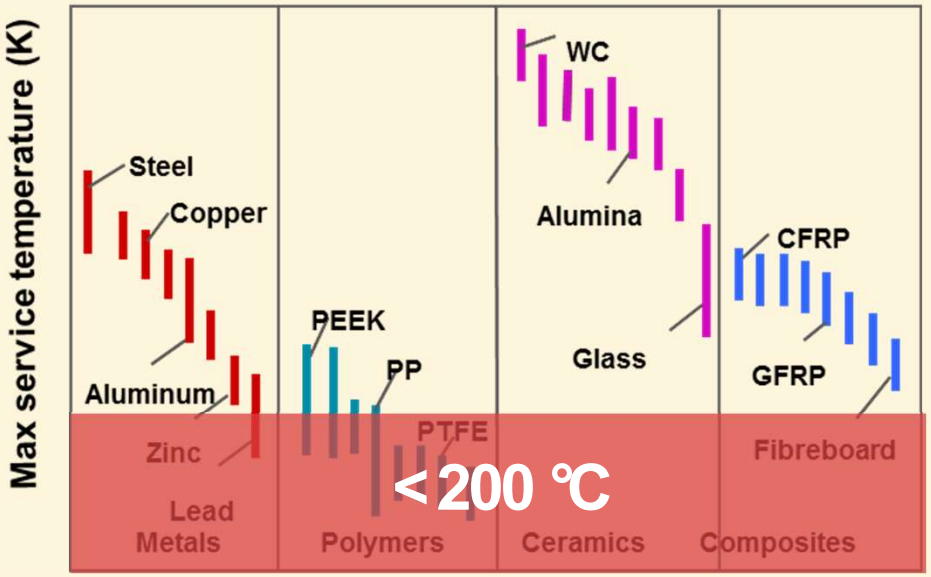
- Operate at 200°C
- Be a good electrical insulator
- Be a good thermal conductor
- Lightweight

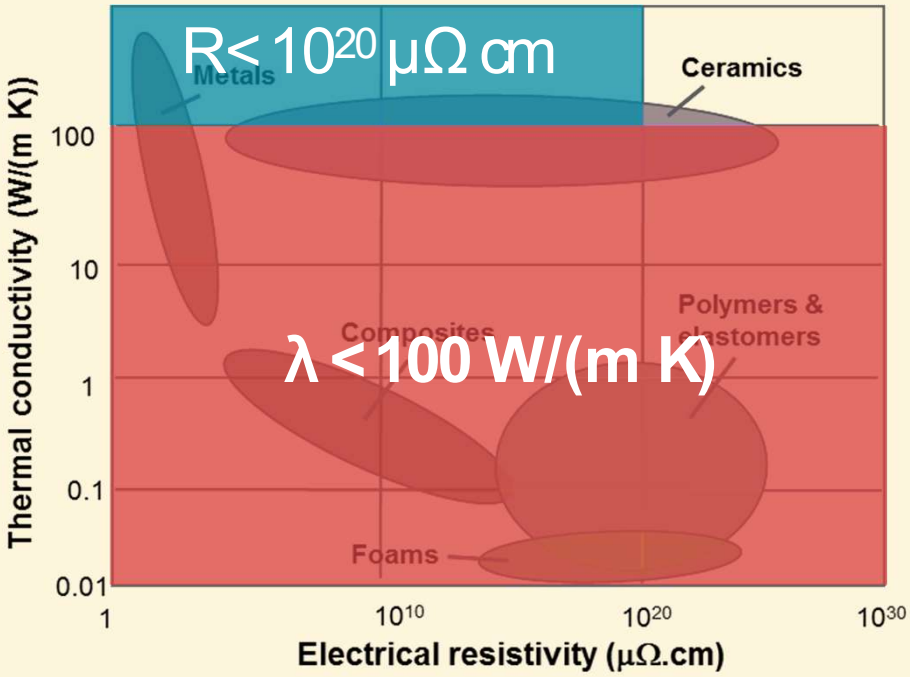
Free variables

- Any suitable material / any process

Objectives

- Max. service temperature > 200°C
- Resistivity > 10^{20} μΩ cm
- Thermal conductivity > 100 W/(m K)
- Density < 4 g/cm^3 (Mg/m^3)





Step 2 – Screening

- There are typically minimum and maximum acceptable limits that define material constraints
- We can therefore rule out materials outside of these…

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Materials selection process (2)

1 – Translation	• Express design requirements as **constraints** and **objectives**
2 – Screening	• **Eliminate materials** based on constraints and objectives
3 – Ranking	• **Rank materials against metrics** describing best performance
4 – Supporting information	• Use additional information to **refine selection**

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Step 3 – Ranking materials

- How can we robustly rank materials for a given design problem?

- Any combination of **function**, **constraints**, **objectives** and **free variables** leads to a **material performance index**

- These can be used to **quantify** the relative performance of one material against another for a given design problem

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Ranking by indices

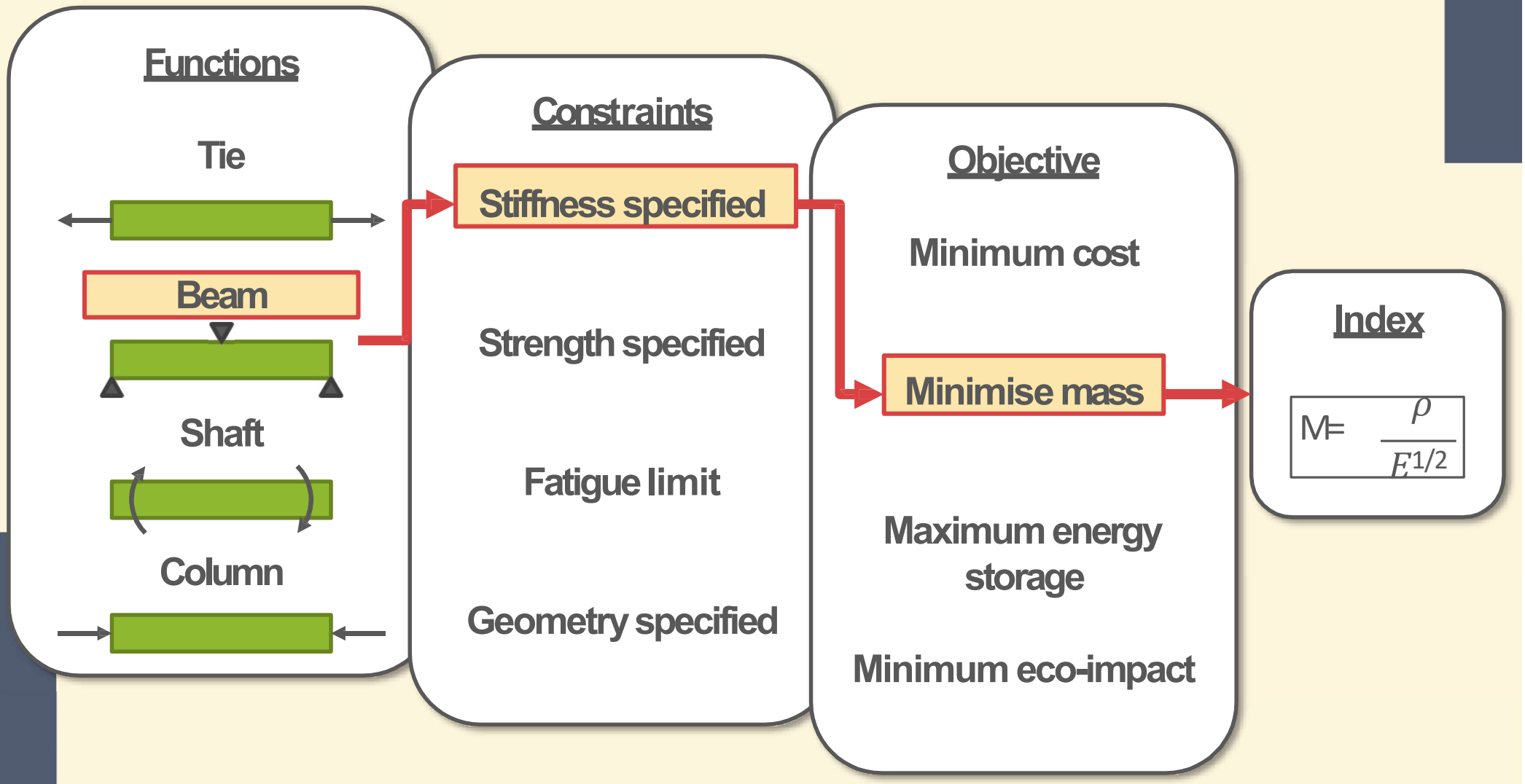
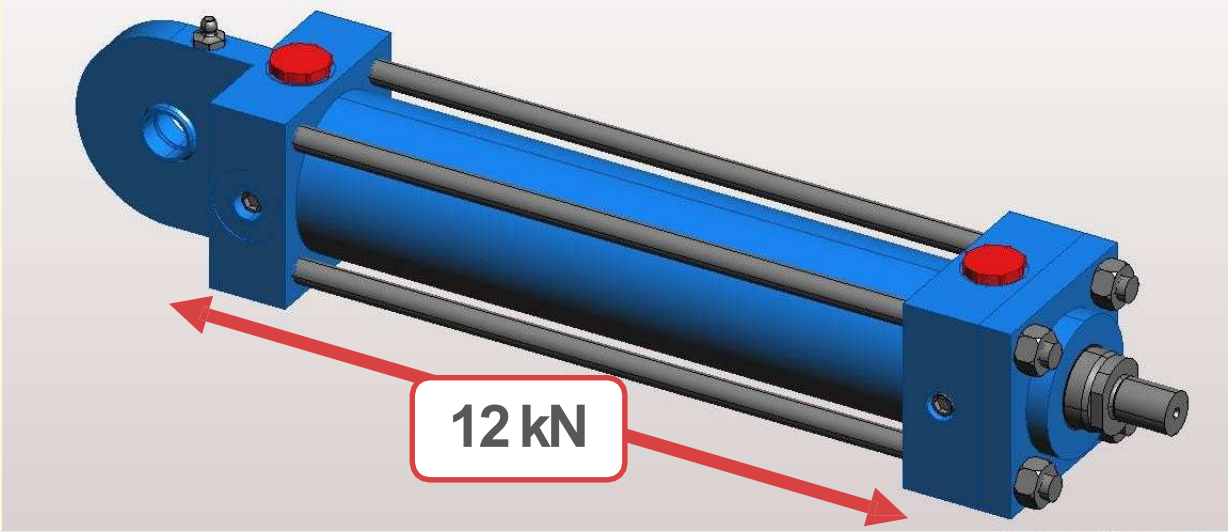


Image adapted from: Ashby, M.F., (2011), *Materials Selection in Mechanical Design*

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Double-acting hydraulic cylinder
Thornton, 2006



Example – Hydraulic cylinder tie rods (1)

- In high-pressure cylinders, high strength steel tie rods carry the axial load generated by the internal pressure – this can be very high in industrial systems!

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Example – Hydraulic cylinder tie rods (2)

- Constraints:
  - Sustain axial load (12 kN) resulting from cylinder pressure (**strong**)
  - Fit connections on existing end blocks and cylinders
  - Required for large scale industrial use (**low cost**)
- Free variables:
  - **Any material** made by **any manufacturing process**
  - Tie rods can be of **any diameter** (within reason!)
- Objectives:
  - Sustain 12 kN load without **failure by yield or fracture**
  - Maintain threaded and bolted connections to end blocks
  - **Minimise cost**

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Example – Hydraulic cylinder tie rods (3)

Failure function

- Avoid failure by yielding or fast fracture
- Both depend on staying below a 'critical' stress
- Failure strength (σ_f) must be greater than applied stress:

$$-\frac{F}{A} < \sigma_f \qquad (1)$$

Objective function

- Assume cost solely based on material used
- Therefore we need to minimise mass (m):
  - $m = AL\rho$
- Considering material cost per unit mass (C_m):
  - $C = C_m\ m = C_m\ AL\rho \qquad (2)$

The free variable (A) can now be used to combine functions.

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Example – Hydraulic cylinder tie rods (4)

- Failure function: $\frac{F}{A} < \sigma_f$ (1)
- Objective function: $C = C_m A L \rho$ (2)

- Combine by using the **free variable** (A):
  - $C = C_m A L \rho =$

- Performance index is based on **material properties**:
  - C=

We want to minimise cost and density, while maximising strength

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Data from: Ashby, M. F., (2011), *Materials Selection in Mechanical Design*

Material	Cost (£ kg^{-1})	Density (kg m^{-3})	Failure strength (MPa)	Index, *M*
Medium carbon steels	0.44 – 0.48	7800 – 7900	305 – 900 (yield)	
Aluminium alloys	0.98 – 1.11	2500 – 2900	30 – 500 (yield)	
Carbon-fibre reinforced plastic	26 – 29	1500 – 1600	550 – 1050 (tensile)	
Bamboo	0.91 – 1.37	600 – 800	35 – 44 (tensile)	

Example – Hydraulic cylinder tie rods (5)

- Material performance index (**smaller is better**):

– M =

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Step 4 – Refining selection

- Steps 1 to 3 do not usually produce a single option for a material to use
- Need **additional information** to refine the short-list and make a final design choice:
  - **Reputation** (i.e. of material consistency, producer or supplier)
  - **Precedents** (i.e. has it been used for a similar application before?)
  - **Availability** (i.e. can it be sourced in the required volumes / time?)
- This additional information can take many forms:
  - **Case studies** (e.g. articles, journals, reports)
  - **Market data** (e.g. material price changes)
  - **Social-political-economic** circumstances!

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Summary

- To enable us to select the 'best' material for a particular application, we need to apply a structured **selection methodology**

- The 'best' material can be defined by a number of factors:
  - **Physical** (strength, stiffness, toughness etc.)
  - **Economic** (cost, manufacturing considerations etc.)
  - **Political** (availability – who own it)

- By **translating** our design requirements, **screening**, **ranking** and **refining** our materials choice, we can usually identify a suitable candidate for a particular application

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Concepts and terminology

- Materials Selection:
  - **Translating**
  - **Screening**
  - **Ranking**
  - **Refining**
- Translating:
  - **Functions**
  - **Constraints**
  - **Free variables**
  - **Objectives**
- Ranking:
  - **Material Performance Index**
  - **Failure function**
  - **Objective function**
- Refining:
  - **Additional information**